# BREAST CARE-MA DIAGNOSTIC DECISION SUPPORT REPORT

Department of Information Technology | Nehru Institute of Engineering and Technology
Explainable Multi-Agent AI System for Ultrasound Modality

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| **Case ID:** | BUSI-BENIGN-911C | **Date & Time:** | 2026-10-31 00:11:47 |
| **Patient Age:** | 52 | **Family History:** | No |
| **Breast Density:** | C - Heterogeneously dense | **Symptoms:** | Palpable nodule |

## 1. Quantitative AI Findings & Model Outputs

| Parameter | Result | Clinical Meaning |
|---|---|---|
| AI Classification | **NORMAL** | Calibrated Malignant Probability: 8.2% |
| Tumor Detection | Lesion Detected | Relative Area: 37.10% |
| Morphology (Agent 3) | Irregular / Spiculated | Aspect Ratio: 1.32, Solidity: 0.63 |
| Acoustic Shadowing | Absent | Acoustic attenuation distal to mass |
| ACR BI-RADS Category | **BI-RADS 1** | **Negative / Routine Screening** |

## 2. Retrieved Evidence & Clinical Guidelines (Agent 4)

• **Normal Ultrasound Examination** (ACR BI-RADS Atlas 5th Edition (Ultrasound)): No masses, architectural distortion, or suspicious calcifications are detected. Routine age-appropriate screening is recommended.

• **Classic Malignant Presentation (>=95% Probability)** (ACR BI-RADS Atlas 5th Edition / Society of Breast Imaging): Classic presentation: Spiculated margins, taller-than-wide (non-parallel) orientation, hypoechoic with acoustic shadowing, microcalcifications. Malignancy risk exceeds 95%.

## 3. Recommended Clinical Follow-up Actions

• Routine annual screening mammography/ultrasound as per age guidelines.

• Clinical breast examination (CBE) during annual wellness visit.

**REGULATORY & SAFETY DISCLAIMER:** NOTICE: Breast CARE-MA is an AI decision-support research prototype developed for academic evaluation. It does not provide definitive medical diagnosis and must never supersede clinical judgment by a licensed radiologist or oncologist.